

Folic acid-functionalized gadolinium-loaded phase transition nanodroplets for dual-modal ultrasound/magnetic resonance imaging of hepatocellular carcinoma

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ABSTRACT

Dual-modal molecular imaging by combining two imaging techniques can provide complementary information for early cancer diagnosis and therapeutic monitoring. In the present manuscript, folic acid (FA)-functionalized gadolinium-loaded nanodroplets (NDs) are introduced as dual-modal ultrasound (US)/magnetic resonance (MR) imaging contrast agents. These phase-change contrast agents (PCCAs) with alginate (Alg) stabilizing shell and a liquid perfluorohexane (PFH) core were successfully synthesized via the nano-emulsion method and characterized. In this regard, mouse hepatocellular carcinoma (Hepa1-6) as target cancer cells and mouse fibroblast (L929) as control cells were used. The *in vitro* and *in vivo* cytotoxicity assessments indicated that Gd/PFH@Alg and Gd/PFH@Alg-FA nanodroplets are highly biocompatible. Gd-loaded NDs do not induce organ toxicity, and no significant hemolytic activity in human red blood cells is observed. Additionally, nanodroplets exhibited strong ultrasound signal intensities as well as T₁-weighted MRI signal enhancement with a high relaxivity value of 6.40 mM⁻¹ s⁻¹, which is significantly higher than that of the clinical Gadovist contrast agent (r₁ = 4.01 mM⁻¹ s⁻¹). Cellular uptake of Gd-NDs-FA by Hepa1-6 cancer cells was approximately 2.5-fold higher than that of Gd-NDs after 12 h incubation. Furthermore, *in vivo* results confirmed that the Gd-NDs-FA bound selectively to cancer cells and were accumulated in the tumor region. In conclusion, Gd/PFH@Alg-FA nanodroplets have great potential as US/MR dual-modal imaging nanoprobes for the early diagnosis of cancer.

1. Introduction

Molecular imaging has recently shown considerable potential for early cancer diagnosis and the monitoring of therapeutic techniques [1]. Several non-invasive molecular imaging methods, including Ultrasound Imaging (USI), Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and/or Fluorescence Imaging (FI), are currently used for clinical cancer diagnosis and management [2,3]. Although these techniques offer many advantages, each one suffers from intrinsic limitations such as low specificity, sensitivity, spatial resolution, and tissue penetration depth in cancer diagnosis [3–5]. In recent years, multimodal

imaging techniques have shown a unique superiority in combination advantages of two or more imaging techniques and provide complementary information for more accurate and reliable cancer diagnosis and therapeutic monitoring [3,6–8]. For example, MRI with high spatial resolution, excellent soft-tissue contrast, deep tissue penetration, and multi-planar imaging suffers from low sensitivity [3,5,7,9,10]. Conversely, USI is a cost-effective portable clinical diagnostic imaging technique that provides real-time images with high sensitivity but low spatial resolution [3,9]. Furthermore, progresses in nanotechnology have facilitated the integration of USI and MRI techniques using a single contrast agent (CA). US/MR imaging dual contrast agents provide

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comprehensive and accurate diagnostic information for early-stage cancer diagnosis and image-guided therapies [6,7].

Microbubbles (MBs) have been widely used as ultrasound CAs for many decades and have been proven to be highly efficient US molecular probes in therapeutic applications, including drug and gene delivery [11,12]. MBs in response to ultrasound waves oscillate and provide non-linear signals owing to a mismatch in acoustic impedance between their gas core and surrounding tissues. However, they suffer from some limitations such as poor stability, short circulation time, and relatively large size [13,14]. Therefore, MBs cannot effectively penetrate tumor tissue, and their applications are limited to intravascular space imaging [11,15,16]. To overcome these shortcomings, phase-change contrast agents (PCCAs) such as nanodroplets (NDs) were developed for US molecular imaging [12,17,18]. PCCAs typically have a liquid core of low-boiling point perfluorocarbon (PFC) like perfluorohexane (PFH) with a stabilizing shell that can vaporize to the gas phase upon activation by ultrasound energy in an event termed acoustic droplet vaporization (ADV) [17–20]. The stability of PFH liquid NDs facilitates long circulation time *in vivo*, and their nano-scaled size enables them to effectively penetrate and accumulate in tumor tissues via the enhanced permeability and retention (EPR) effect. These acoustic phase-changeable NDs enhance ultrasound image contrast by vaporizing into echogenic MBs [12,17,21,22]. These favorable properties of PFH nanodroplets make them promising candidates for biomedical applications such as ultrasound imaging and therapy [18].

Various shells such as lipid, polymer, and protein can be used to enhance the bio-stability of NDs in the physiological environment by reducing the surface tension and providing a diffusion barrier. Moreover, stabilizing shells can functionalize the surface for the binding of ligands, and loading drugs or contrast agents into NDs [23–26]. Among natural biopolymers, alginate (Alg) has attracted significant attention for coating the NPs in biomedical applications due to its favorable properties such as abundance, non-antigenicity, high biocompatibility, good biodegradability, and hydrophilic nature [27–29].

Gadolinium (Gd) is the most common metal ion as an MRI contrast agent owing to its inherent excellent physicochemical properties and strong contrast efficiency [30–32]. It has seven unpaired electron spins that significantly enhance r_1 relaxivity [32,33]. However, free Gd ions (Gd^{3+}) induce neurologic and cardiovascular toxicity because of inhibiting calcium channels and causing changes in cellular reactive oxygen species (ROS) levels [34–36]. Currently, paramagnetic Gd-based complexes or chelating compounds with high biocompatibility and appropriate clinical effects have been developed as MRI contrast agents [1, 37]. The clinically approved gadolinium chelates such as Gd-DO3A-butrol (Gadovist) mainly consist of central Gd^{3+} ion with chelating ligands [32]. However, owing to the release of some Gd^{3+} *in vivo* applications, severe side effects such as long-term deposition in brain tissue and nephrogenic systemic fibrosis (NSF) occurred [33,38]. One approach for reducing biological toxicity is loading these agents into nanocarriers such as inorganic nanoparticles, micelles, microcapsules, liposomes, and sonosensitive NDs. These nano contrast agents without the leakage of toxic Gd^{3+} ions and loss of prominent MR relaxivity indicate high selectivity, stability, and biosafety [31,39].

Nanoparticles (NPs) tend to passively penetrate tumor sites using the EPR effect via extravasation through endothelial cells from the leaky vasculature [40,41]. However, passive targeting is inefficient owing to some limitations such as low specific binding of NPs with the cancer cells, a strong dependence of targeting efficiency on the physicochemical properties of NPs (e.g., size, shape, etc.), permeability, and variety of the degree of vascularization in tumor types [40,42]. Surface functionalization of NPs with various specific bioligands such as folic acid (FA) [43,44], antibodies [45,46], aptamers [47,48], and peptides [49, 50], which are overexpressed on certain cancer cells, can enhance the targeting specificity and cellular uptake of NPs by target cells [51]. This strategy, known as active targeting, can improve the accuracy of cancer cell detection [52]. FA is a non-immunogenic, biocompatible,

inexpensive, and highly stable small molecule ligand with a high specific binding affinity to folate receptors (FRs) on cancer cells. This attractive targeting ligand has been widely used for bio-conjugation of NPs to enhance the diagnostic and therapeutic efficiency of many cancers, including hepatocellular carcinoma (HCC) [44,53–55].

The present study aimed to develop ultrasound-triggered phase transition NDs with dual-modal US/MR imaging functionality. Most dual-modal ultrasound/MRI micro or nanoprobe apply T_2 -weighted MR imaging contrast agents such as Fe_3O_4 [6,9,10,56], superparamagnetic iron oxide nanoparticles (SPIONs) [4,5,57], Fe-doped silica [45], and hollow mesoporous prussian blue nanoparticles (HMPBs) [58]. Since T_2 -weighted MR imaging suffers from hypointense lesions and the blooming effect, many efforts have been made to develop T_1 -weighted contrast agents in recent decades. Gd-based T_1 contrast agents that produce brighter images can be more easily detected against image backgrounds [59]. Herein, we developed Gd-loaded NDs with liquid PFH cores and alginate/surfactant shells that produce positive contrast in T_1 -weighted MR images. These nanodroplets can be triggered by clinical ultrasound waves and can be vaporized to echogenic MBs to enhance ultrasound and MR imaging contrast. The NDs were functionalized with FA to enhance their tumor-targeting ability. To the best of our knowledge, Gadovist which is commonly used in clinical MRI has not yet been loaded into alginate-coated and folic acid-targeted perfluorohexane nanodroplets to investigate the diagnostic potential of this system in ultrasound/MR dual-modal imaging of cancers. The synthesized targeted nano-scaled CAs in this study with superior acoustic and magnetic properties can be specifically delivered to the target tumor and provide higher US/MR imaging contrast than clinical available CAs.

2. Material and methods

2.1. Materials

Perfluorohexane (PFH, 99%), Folic Acid (FA), antibiotic solutions (penicillin-streptomycin), sodium alginate (Alg), 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT), and Dimethyl Sulfoxide (DMSO) were obtained from Sigma-Aldrich (Germany). Phosphate-Buffered Saline (PBS), Surfactant (Tween 20), N-hydroxysuccinimide (NHS), trypsin, 1-ethyl-3-(3-dimethyl aminopropyl) carbodiimide (EDC), and Fetal Bovine Serum (FBS) were obtained from Sigma-Aldrich (USA). Gadobutrol ($C_{18}H_{31}GdN_4O_9$, Gadovist) was obtained from Bayer (Germany). Dulbecco's Modified Eagle Medium (DMEM) was purchased from Biosera Europe, ZI du Bousquet (France). Normal mouse fibroblast (L929) and mouse hepatocellular carcinoma (Hepa1-6) cell lines were purchased from the Pasteur Institute (Tehran, Iran).

2.2. Conjugation of alginate with FA

Folic acid was conjugated with alginate by an EDC/NHS coupling reaction between the amine groups of FA and carboxyl groups of alginate. Alginate, EDC, and NHS with molar ratios of 8:1.5:1 were dissolved in 4 mL of deionized distilled water and were incubated under stirring for 4 h. Subsequently, 20 mg of FA was added to the solution and was further stirred under the dark conditions at room temperature for 12 h.

2.3. Synthesis of non-functionalized and FA-functionalized Gd-NDs

Gd-loaded alginate-stabilized PFH nanodroplets were synthesized using the nano-emulsion method. Briefly, 500 μ L of the PFH solution in 2 mL of distilled deionized water was sonicated using a UP 400S ultrasonic probe for 5 min in an ice bath. Subsequently, 500 μ L of Gadovist and 10 μ L of Tween 20 were added to the solution, followed by sonicating for 4 min. Then, alginate polymer solution (1.5% w/v), including alginate (for Gd-NDs) or alginate-FA (for Gd-NDs-FA) was added dropwise during 5 min sonication to stabilize NDs. Calcium chloride ($CaCl_2$,

0.2% w/v) solution was then added to the emulsion under sonication for 4 min. To complete the process, the obtained solution was magnetically stirred for 2 h at room temperature. Afterward, the resultant emulsion was centrifuged at 11000 rpm at 4 °C for 30 min. Finally, the supernatant of the emulsion was removed, and the deposited NDs were dispersed in 5 ml of PBS. The obtained solution was stored at 4 °C.

2.4. Apparatus and characterizations

The size and size distribution of blank NDs, Gd/PFH@Alg nanodroplets (Gd-NDs), and Gd/PFH@Alg-FA nanodroplets (Gd-NDs-FA) were analyzed by Dynamic Light Scattering (DLS, Vascov/Cordouan Technologies, France). The size and morphology of Gd-NDs-FA were also determined using the Transmission Electron Microscopy (TEM, Philips em208s, 100 kV, Netherlands). Conjugation of FA with Gd-NDs was confirmed by a Fourier Transform Infrared (FTIR) spectrophotometer (JASCO, Japan). The magnetic properties of Gadovist, Gd-NDs, and Gd-NDs-FA were assessed by a vibrating sample magnetometer (VSM, MDKB, Meghnatis Daghigh Kavir, Iran) at room temperature. The loading efficiency and the concentration of gadolinium in NDs emulsions were determined by an inductively coupled plasma optical emission spectrometer (ICP-OES, Varian Vista-Pro, Australia). The stability of Gd-NDs-FA nanodroplets dispersed in PBS buffer at pH = 7.4 was evaluated by monitoring the changes in their hydrodynamic size by DLS, along with four months of storage at 4 °C.

2.5. In vitro ultrasound imaging

The acoustic droplet vaporization (ADV) effect and the potential application of fabricated phase transition NDs as ultrasound contrast agents were evaluated by a clinical diagnostic ultrasound system (UGEO H60, Samsung) equipped with a 12 MHz linear array probe as a signal transmitter and a receiver. PBS was used as a control solution. The images were obtained with the mechanical index (MI) of 1.4 before and after the injection of NDs via a needle into the PBS solution. Additionally, ultrasound imaging of NDs by changing the mechanical index was performed. For this purpose, 1% agarose gel phantom containing 0.5% cellulose was prepared as a standard acoustic scatterer. The sample vial was placed into the gel phantom, and the entire assembly was immersed and fixed into a water bath (Fig. 2(c)). The ultrasound probe was positioned perpendicular to the sample vial into the water bath. The water temperature was maintained at physiological temperature (37 °C) during the imaging test to prevent the thermal inducing phase transition of NDs. Ultrasound images of NDs emulsion were taken with the MI of 0.2, 0.8, and 1.4.

2.6. Relaxivity measurements

To assess the potential application of fabricated Gd-loaded NDs as new MRI nano-CAs, their relaxivity properties, compared to the commercially available CA (Gadovist), were evaluated. Before performing relaxivity measurements, the Gd³⁺ contents in the non-functionalized and FA-functionalized NDs were measured by ICP-OES. The T₁-weighted MR imaging of Gd/PFH@Alg, Gd/PFH@Alg-FA, and Gadovist was performed by a clinical 1.5 T MRI system (Magnetom Aera, Siemens, Germany) using a 32-channel receive-only head coil. For this approach, the samples were diluted in distilled water at various Gd concentrations (0, 0.02, 0.04, 0.08, 0.1, 0.2, 0.4, 0.6, 0.8, and 1 mM) in a series of microtubes and were prepared for MR imaging. T₁-weighted multi spin-echo (SE) imaging sequences of all samples were obtained using the following scanning parameters: Echo time (TE) of 9 ms, variable repetition time (TR) values of 150, 300, 500, 700, 900, 1200, 2000, and 3000 ms, the field of view (FOV) of 192 × 220 mm², slice thickness of 3 mm, flip angle of 90°, and matrix size of 224 × 320. The MRI data were analyzed using an in-house script developed in the MATLAB software (MathWorks, Natick, MA, USA) and the hMRI toolbox (<http://h>

mri.info). In the T₁ mapping procedure, one-parameter models were implemented, and a circular region of interest (ROI) was semi-automatically drawn by the Hough Transform technique for edge detection and segmentation in the MATLAB software. Afterward, voxel-wise T₁ values were computed in the selected ROIs by Eq. (1), and the Levenberg-Marquardt nonlinear least-squares regression was performed on the dataset.

$$SI(\text{voxel}(i,j)) = S_0(\text{voxel}(i,j)) [1 - \exp(-TR/T_1(\text{voxel}(i,j)))] \quad (1)$$

Where SI is the signal intensity of each concentration of the samples, and S₀ shows the signal intensity of water. Then, the mean and standard deviation measures for all voxels within the ROIs were computed. The R₁ relaxation rate (i.e. 1/T₁) was then plotted as a function of different concentrations by Eq. (2).

$$R_1 = 1/T_1 \quad (2)$$

At last, the longitudinal relaxivity (r₁) was calculated by the linear least-squares regression on this plot and the goodness of fit for evaluation using the adjusted R₁ metric.

2.7. Cell culture

Mouse hepatocellular carcinoma (Hepa1-6) and mouse normal fibroblast (L929) cells were cultured in the DMEM culture medium supplemented with 100 µg mL⁻¹ of streptomycin, 100 U mL⁻¹ of penicillin, and 10% FBS. The cultured cells were maintained in the incubator at 37 °C under a 5% CO₂ atmosphere.

2.8. Cell viability assay

The cytotoxicities of targeted and non-targeted Gd-NDs on Hepa1-6 cancer and L929 control cell lines were assessed using the MTT assay. The cells were seeded in 96-well plates (2 × 10⁴ cells/well) and were incubated at 37 °C under a humidified 5% CO₂ atmosphere for 24 h. Then, Gd-NDs and Gd-NDs-FA suspensions at different Gd concentrations (0, 0.1, 0.2, 0.4, 0.6, 0.8, and 1 mM) were added to the cells and then incubated for another 24 h. Afterward, the media containing NDs were removed, and the cells were washed twice with PBS. Subsequently, 100 µL aliquots of culture medium containing 20 µL of the MTT solution (0.5 mg mL⁻¹) were added to the wells, and the plates were maintained under the dark conditions for 4 h. Then, the supernatant was removed, and 150 µL of DMSO was added to each well, and the cells were incubated with shaking for 10 min to dissolve the formazan crystals. Finally, the optical density (OD) value of each sample was measured using a microplate reader (Bio-RAD 680, USA) at a wavelength of 570 nm. The cell viabilities were calculated as percentiles by the ratio of OD values of treated and control groups. Each experiment was repeated three times under identical conditions.

2.9. In vivo toxicity analysis on normal mice

All animal experiments in this study were conducted according to the guidelines for the use of laboratory animals of the Animal Care and Ethics Committee of Isfahan University of Medical Sciences. To evaluate the organ toxicity and blood biochemistry, healthy (tumor-free) mice were divided into two groups (n = 5) and Gd-NDs-FA (250 µL, 0.1 mmol Gd kg⁻¹) was injected intravenously (IV) to the first group. Meanwhile, the same volume of PBS was injected into the control group. After three weeks, the mice were sacrificed, their vital organs, including kidneys, liver, heart, lung, spleen, and brain were harvested, and their blood samples were collected via the cardiac puncture. The blood urea nitrogen (BUN), alanine aminotransferase (ALT), creatinine (CREA), and aspartate aminotransferase (AST) concentrations were measured as hematological analyses using an automatic biochemical analyzer (Biolis 24i Premium Biochemistry Analyzer, Boeki, Japan). Furthermore, the

harvested organs, after washing with PBS, were fixed in formaldehyde (4%) and then embedded in paraffin. Subsequently, the organs were dehydrated, blocked, sectioned at an approximate thickness of 5 μm , and then stained with hematoxylin and eosin (H&E) for the histological exam. Finally, histological photographs of organ sections were recorded using a digital light microscope (Olympus, Tokyo, Japan).

2.10. Hemolysis assay on human blood

To evaluate the hemocompatibility of Gadovist and Gd-NDs, fresh human blood samples were collected from healthy volunteers. To harvest red blood cells (RBCs), blood samples were centrifuged at 1500 rpm for 15 min. Then, the RBCs sediments were washed three times with sterile normal saline by centrifugation at 1500 rpm for 25 min and were diluted in normal saline. Subsequently, the RBCs suspensions (200 μL) were mixed with different solutions (800 μL), including Gadovist, blank NDs and Gd-NDs-FA with different Gd concentrations (0.07, 0.15, 0.31, 0.62, 1.25, 2.5, 5, and 10 mM). The positive control (complete hemolysis, 100%) and the negative control (no hemolysis, 0%) were obtained by mixing the RBCs suspension (200 μL) with the Triton X100 (800 μL) as well as the PBS buffer (800 μL), respectively. Subsequently, all samples were gently shaken and incubated at 37 $^{\circ}\text{C}$ for 2 h, and then were centrifuged at 11000 rpm at 4 $^{\circ}\text{C}$ for 30 min. The supernatants were added to a 96-well culture plate, and then the absorbance value of each sample, related to the concentration of released hemoglobin, was measured using a UV-vis spectrophotometer (Shimadzu UV-160, Japan) at 540 nm. Finally, the hemolysis percentage of the samples was calculated using the following equation:

$$\text{Hemolysis \%} = (\text{Abs}_{\text{Sample}} - \text{Abs}_{\text{Ctrl-}}) / (\text{Abs}_{\text{Ctrl+}} - \text{Abs}_{\text{Ctrl-}}) \times 100 \quad (3)$$

Where, $\text{Abs}_{\text{sample}}$ is the absorbance of treated cells with Gadovist or NDs, and $\text{Abs}_{\text{Ctrl+}}$ and $\text{Abs}_{\text{Ctrl-}}$ show the absorbance of the positive and negative controls, respectively.

2.11. Intracellular uptake assay of NDs

The amounts of internalized Gadovist and Gd-loaded NDs into the Hepa1-6 cells were determined by the ICP-OES assay. The cells were cultivated in 6-well plates (2×10^6 cells per well) and were incubated at 37 $^{\circ}\text{C}$ under 5% CO_2 atmosphere for 24 h. After that, the cells were treated with Gadovist, Gd-NDs, and Gd-NDs-FA nanodroplets at 0.8 mM Gd concentration and were incubated for 2, 6, 12, 18, and 24 h. Then, the supernatant was removed, the cells were washed three times with PBS, and they were harvested by trypsin and counted. Subsequently, they were centrifuged and dissolved in 3 mL aqua regia solution (3:1 (v/v)) HCL/ HNO_3 to lyse the cells. Finally, the Gd^{3+} uptake per cell was quantified by ICP-OES. All measurements were performed in triplicate, and the mean $\pm$ SD of the results was calculated.

2.12. In vitro MR imaging

T_1 -weighted MR imaging experiments on Hepa1-6 cancer and L929 normal cell lines were performed by a clinical 1.5 T S Aera MRI Scanner. For these experiments, the cells were seeded in 6-well plates (3×10^6 cells/well), and they were maintained overnight in a 5% CO_2 atmosphere at 37 $^{\circ}\text{C}$. Then, the cells were incubated with Gd-NDs and Gd-NDs-FA nanodroplets with different Gd^{3+} ion concentrations of 0, 25, 50, and 100 $\mu\text{g mL}^{-1}$ for 6 h. Subsequently, they were washed three times with PBS, detached using trypsin, and suspended in 2% agarose gel. Finally, the T_1 -weighted MRI images were recorded using the following scanning parameters: flip angle = 90 $^{\circ}$, TE = 9 ms, TR = 500, matrix size = 224×320 , slice thickness = 3 mm, and FOV = $192 \times 220 \text{ mm}^2$. To quantify the MRI images, the average of voxel intensities of micro-tubes was calculated as signal changes based on different Gd^{3+} concentrations and the structure of nanoprobe. All data were

subsequently analyzed using the MATLAB 9.4 software, and statistical evaluations were performed using the IBM SPSS Statistics software (version 20.0.0, IBM Corp, Armonk, NY, USA).

2.13. In vivo ultrasound imaging

The animal tumor-bearing model was established on 5-6-week-old C57BL/6 (18–22 g) mice using subcutaneously inoculating Hepa1-6 cell line. The 5 tumor-bearing mice were subjected to ultrasound imaging when the size of the tumor reached 2–3 cm. Afterward, 250 μL of Gd-NDs-FA nanodroplets emulsion was intravenously injected via the tail vein of the mice, and the ultrasound probe was simultaneously placed in the tumor region. The space between the probe and the tumor surface was filled with sufficient ultrasonic gel. Ultrasound images were acquired by a portable ultrasound system (SIUI, CTS-900V, China) using a linear array transducer at the frequency of 7.5 MHz and MI of <0.83 before and after contrast injection at different times (10, 20, and 60 min). The imaging parameters were constant during the entire imaging experiment session. The contrast enhancement efficiency of the NDs in the tumor was evaluated by calculating the signal intensity of US images within the ROI located in the tumor region.

2.14. In vivo MR imaging

For *in vivo* MR imaging, 5 mice bearing Hepa1-6 tumor were anesthetized by injecting 100 and 10 mg/kg of ketamine and xylazine, respectively [38]. Then, 250 μL of Gd-NDs-FA emulsion (0.1 mmol Gd kg^{-1}) [60] was intravenously injected into the tail vein of the mice, and T_1 -weighted MR images were acquired by a 1.5 T S Aera MRI Scanner using the head coil before and 1, 6 and 24 h after injection of NDs. The scanning parameters included sequences in the coronal plane, TE: 12 ms, TR: 500 ms, slice thickness: 1.5 mm, matrix size: 192×154 , FOV: 110 mm, and flip angle = 90 $^{\circ}$. The MRI average signal intensity of tumors at each time was calculated by drawing an ROI on the tumor region using the MATLAB software.

2.15. Biodistribution of Gd-NDs-FA in vivo

To evaluate *in vivo* biodistribution of Gd-loaded NDs, Gd-NDs-FA solution (0.1 mmol Gd kg^{-1}) was injected into the Hepa1-6 tumor-bearing mice via the tail vein. The mice were sacrificed 24 h after injection and their tumor, liver, kidney, heart, spleen, and lung were harvested. To measure the Gd^{3+} content in the tumor and different organs, 0.1 g of each organ was homogenized and a solution containing hydrochloric acid (12.5 M) and nitric acid (5 M) was added to the samples. Then, the samples were incubated at room temperature until they were fully digested. Subsequently, Gd^{3+} concentration in each sample was measured using ICP-OES.

2.16. Statistical analysis

Data were analyzed using SPSS software (version 18) and were presented as mean values $\pm$ SD of triplicate experiments. Statistical analyses were performed using the ANOVA and *t*-test method. *p*-value < 0.05 and *p*-value > 0.05 were considered statistically significant and insignificant, respectively.

3. Results and discussion

3.1. Synthesis and characterization of functionalized and non-functionalized NDs

In this study, ultrasound-triggered phase transition Gd-loaded NDs with dual-modal US/MR imaging functionality were prepared via the nano-emulsion process. Alginate, a biocompatible and biodegradable natural polymer, was used as a stabilizing shell of the PFH liquid core. It

can functionalize the NDs surface for binding of FA as an active targeting ligand to enhance the specificity and sensitivity of cancer detection. The presence of surfactant (Tween 20) in the synthesis causes the formation of smaller and higher stable droplets during the emulsification process [22,61]. The prepared NDs convert to echogenic MBs and nanobubbles (NBs) under ultrasound exposure that enhance ultrasound contrast, and loaded Gd^{3+} ions act as MRI contrast agents. The results suggest that synthesized Gd-loaded nanodroplets with excellent properties are promising candidates as dual-modal US/MR imaging CAs for the early detection of cancer.

The size, size distribution, and morphology of the fabricated NDs were assessed by TEM and DLS. The DLS analysis demonstrates that the hydrodynamic (HD) sizes of the blank NDs, Gd/PFH@Alg nanodroplets (Gd-NDs), and Gd/PFH@Alg-FA nanodroplets (Gd-NDs-FA) are 23.4 ± 1.2 , 31.2 ± 1.5 , and 33.5 ± 1.9 nm, respectively. The results indicate that the HD size of NDs after Gadovist loading is increased. Additionally, the average size of Gd-NDs-FA is slightly larger than that of Gd-NDs. The morphology of Gd-NDs-FA nanodroplets in the TEM image (Fig. 1(a)) shows approximately spherical shapes with core-shell structures including dark contents, demonstrating the loading of gadolinium ions within NDs. As shown in the TEM image, some NDs are expanded and their surface is broken. This could be attributed to the conversion of the PFH liquid core to gas under a high vacuum in the TEM analysis [62]. The TEM image shows that NDs have a relatively uniform particle size with an approximate mean size of 26.3 ± 3.1 nm. The HD size of Gd-NDs-FA in DLS is larger than the TEM image size, since the DLS shows the hydrodynamic diameter and water-swollen NPs in a hydrated state, whereas TEM represents the actual NPs size in a dried state of the sample. The small size of fabricated NDs offers high stability and permeability, which is essential for molecular imaging [12].

Storage stability is a crucial factor for imaging CAs. To evaluate the stability of NDs, changes in the HD size of Gd-NDs-FA were analyzed by DLS after incubation in PBS (pH, 7.4) at different times (20, 40, 60, 80, 100, and 120 days) (Supplementary Fig.S1) in triplicate. As Fig. 1(b) shows, no significant changes in the size of NDs (p -value > 0.05) are observed during four months of storage at 4°C , demonstrating their excellent physical stability. The high stability of NDs could be attributed to the stabilizing alginate shell which prevents the breakdown of NDs or PFH gas diffusion of the core [5,14,63]. ICP-OES experiments show that $43.2 \pm 4.9\%$ of the total used Gd^{3+} has been loaded into NDs.

The functionalization of Gd-loaded NDs with FA was confirmed using the FTIR analysis (Fig. 1(c)). In the IR spectrum of FA, an absorption around 1690 cm^{-1} is attributed to the $\text{C}=\text{O}$ stretch vibration. Also, the absorption peaks within the range over $3300\text{--}3500\text{ cm}^{-1}$ are attributed to N-H and O-H vibrations. Furthermore, the stretching absorption peak of C-O appears within the $1350\text{--}1400\text{ cm}^{-1}$ range. Characteristic peaks related to both FA and Gd-NDs can be seen, particularly in the range of $500\text{--}1600\text{ cm}^{-1}$ in the Gd-NDs-FA spectrum. Owing to the removal of the unreacted FA from prepared NDs by centrifugation, spectral similarities confirm the conjugation of FA with Gd-NDs.

3.2. In vitro ultrasound imaging

The potential application of synthesized NDs as ultrasound CAs was evaluated by US imaging. As Fig. 2(a) shows, the image of PBS, as a control solution before the injection of NDs, appears black. NDs, after injection into PBS, change the phase to echogenic MBs as a result of the ADV effect under US exposure. Due to the mismatch in acoustic impedances with surrounding media, the bubbles create US signal echoes and enhance the image contrast [64]. Therefore, they appear as bright

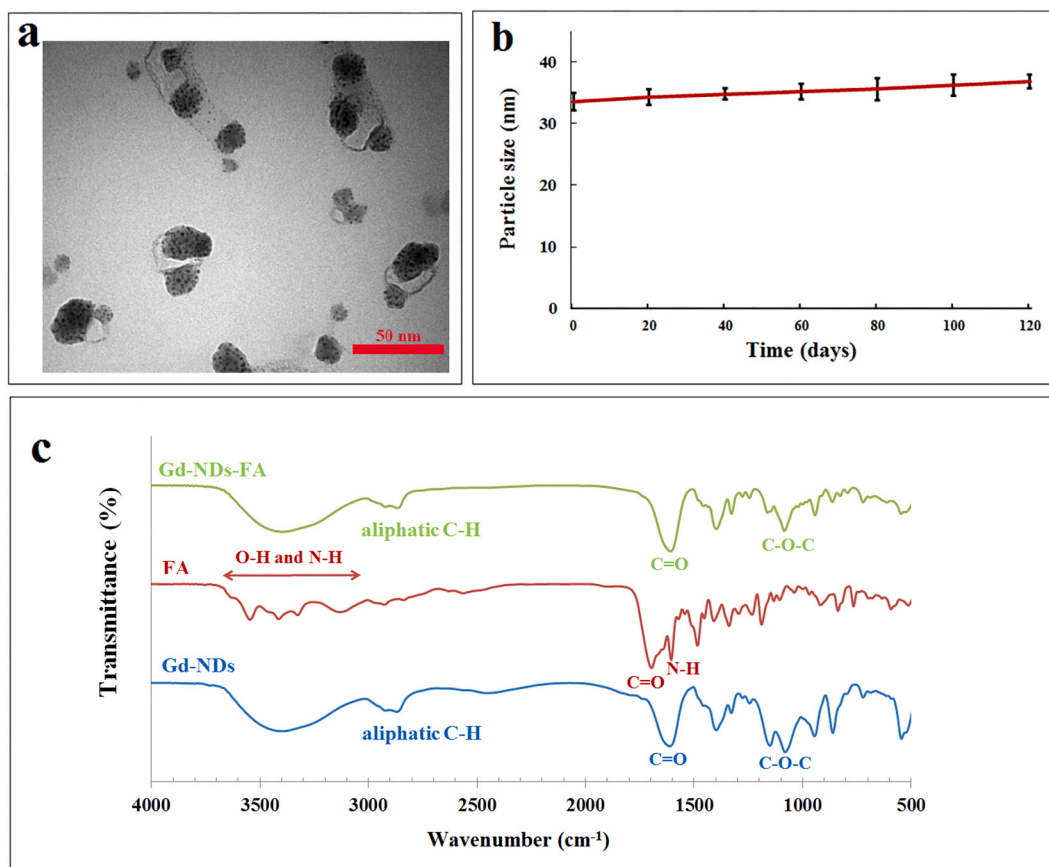


Fig. 1. (a) TEM image of Gd-NDs-FA nanodroplets. (b) Stability evaluation of NDs dispersed in PBS at pH = 7.4 by analyzing the changes in the particle sizes by DLS after the storage at 4°C for four months; Means $\pm$ SD ($n = 3$). (c) FTIR spectra of folic acid (FA), Gd-NDs, and Gd-NDs-FA nanodroplets.

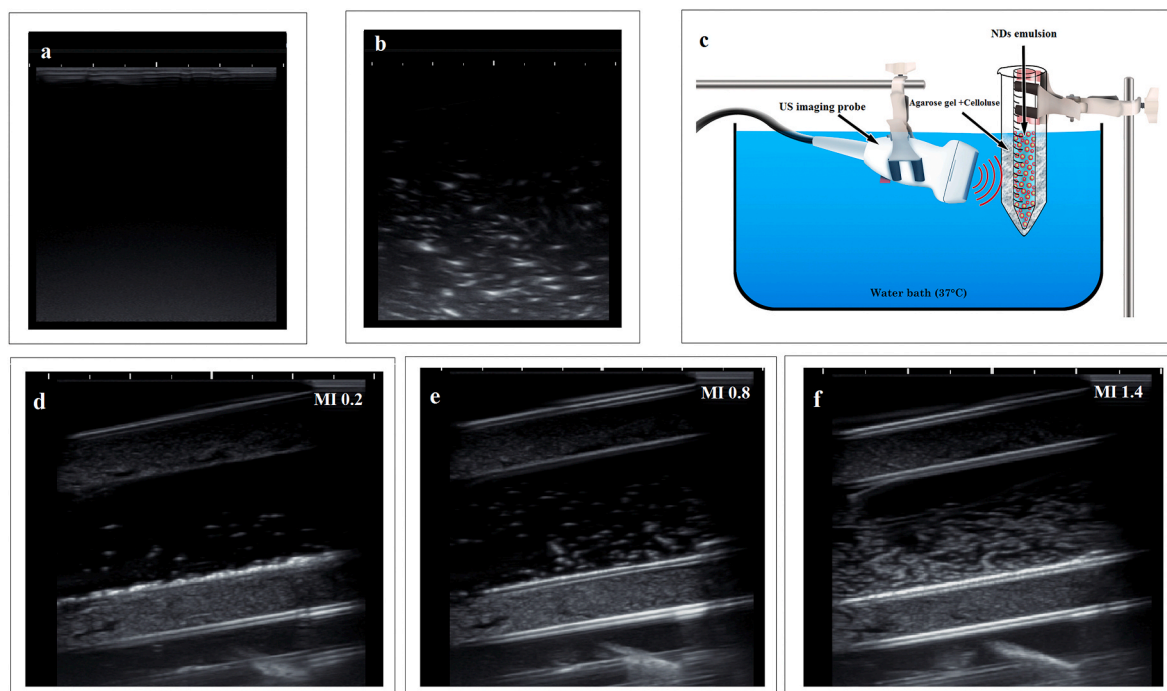


Fig. 2. Ultrasound images of PBS as a control solution (a), NDs after injection into PBS (b), (c) Schematic diagram of the ultrasound imaging setup of NDs at different ultrasound MIs (d–f).

spots in the US image (Fig. 2(b)) that confirm the phase transition of NDs to MBs. Moreover, the echogenic behavior and contrast enhancement ability of NDs were investigated at different ultrasound mechanical index (Fig. 2(c)). For this purpose, samples were exposed to ultrasound with MI from 0.2 to 1.4. As Fig. 2(d) shows, gas bubbles are triggered to be formed under US exposure at the MI of 0.2 that are observed as bright spots in the image. The gas bubble release is gradually increased with

increasing MI, and undergoes a burst release at MI = 1.4 since MI is proportional to the output power of US exposure [26].

3.3. Analysis of magnetic properties of Gd-loaded NDs

The magnetic characterization of Gadovist and Gd-loaded NDs was performed by the VSM analysis. The linear relationship between

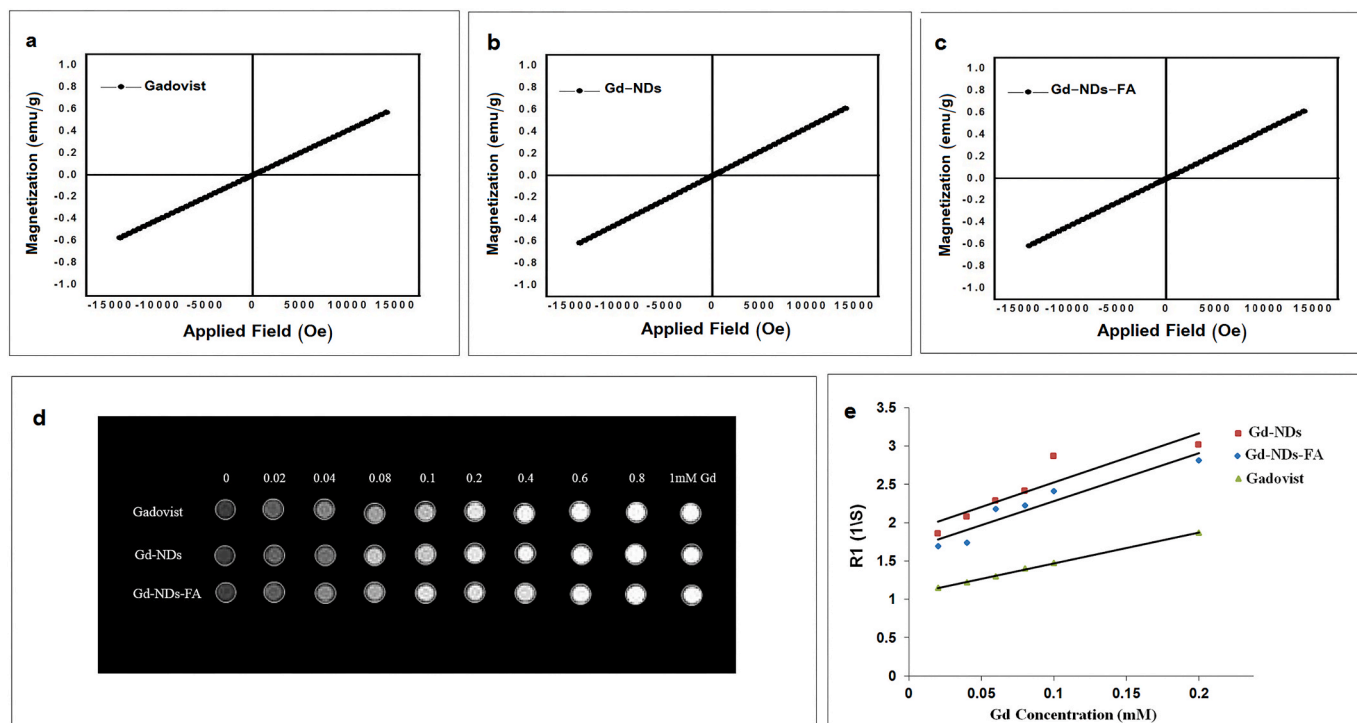


Fig. 3. VSM analyses of Gadovist (a), Gd-NDs (b), and Gd-NDs-FA (c) (magnetization (emu g^{-1}) as a function of applied fields (Oe)) at room temperature. (d) T_1 -weighted MRI images and (e) r_1 relaxivity plots of Gadovist, Gd-NDs, and Gd-NDs-FA at various Gd concentrations.

magnetization and the applied field with a positive slope (Fig. 3(a–c)) confirms the paramagnetic behavior of all samples [65,66]. The saturation magnetization (M_s) values of Gadovist, Gd/PFH@Alg, and Gd/PFH@Alg-FA nanodroplets are 0.57, 0.64, and 0.61 emu g^{-1} , respectively. The saturation magnetization of Gd-NDs after conjugation with FA does not change significantly. The VSM results suggest that fabricated Gd-loaded NDs with excellent paramagnetic properties have great potential as MRI contrast agents.

3.4. Relaxivity measurements

T_1 -weighted MRI of Gd/PFH@Alg and Gd/PFH@Alg-FA nanodroplets, compared to the commercially available MRI contrast agent (Gadovist) at different Gd concentrations, was performed using a 1.5 T MRI human clinical scanner. As Fig. 3(d) shows, the MRI signals exhibit Gd concentration-dependence with positive signal enhancement as the concentration of Gd is increased. Therefore, the relaxation rates (R_1) show an excellent linear relationship with the Gd concentration for all samples (Fig. 3(e)). According to the slope of curves, the r_1 relaxivities of Gd-NDs and Gd-NDs-FA are obtained 6.40 and 6.27 $\text{mM}^{-1}\text{s}^{-1}$, respectively, which are higher than that of Gadovist ($r_1 = 4.01 \text{ mM}^{-1}\text{s}^{-1}$), recently reported Gd_2O_3 nanoparticles ($r_1 = 3.95$) [38], and Gd-doped carbon dots ($r_1 = 6.06$) [59]. According to a previous study, when Gd^{3+} ions are confined in a limited space, their relaxivity could be increased [67]. Since the amount of Gd^{3+} ions per volume and local Gd concentration within NDs were higher than that of Gadovist at the same concentration, Gd-NDs further enhance the MRI relaxivity.

3.5. Cytotoxicity assessment of NDs on cancer and normal cells

The toxicity effects of FA-functionalized and non-functionalized Gd-NDs on L929 and Hepa1-6 cells were evaluated by the MTT assay. The MTT results demonstrated that the cell viabilities of all groups were decreased with a Gd concentration-dependent trend (Fig. 4). The cytotoxicity of Gd-NDs-FA in all concentrations is higher than that of non-functionalized Gd-NDs for the Hepa1-6 cell line, which can be attributed to the overexpression of FA receptors on the surface of cancer cells and the enhancement of cellular internalization of nanoprobe. For instance, the viability percentages of Hepa1-6 cells treated with Gd-NDs and Gd-NDs-FA are 86.1% and 81.3%, respectively at 1 mM Gd concentration, and Hepa1-6 cells show cell viabilities above 80% in all concentrations. Furthermore, high viability rates are observed for L929 cells after 24 h incubation even at the highest concentration (1 mM Gd) so that for both samples, the viability percentages of this cell line are above 90% at all concentrations. Moreover, no significant difference is observed in the viabilities of normal cells treated with non-functionalized and FA-functionalized Gd-NDs ($p\text{-value} > 0.05$), confirming that fabricated nano-contrast agents have excellent biocompatibility for normal cells. Consequently, these results suggest that alginate stabilized Gd-loaded NDs are highly biocompatible and safe, which could be ideal candidates for various biomedical applications, including molecular imaging.

3.6. Histopathology and blood biochemistry exams on normal mice

The main limitations of nanomaterial-based CAs for *in vivo*

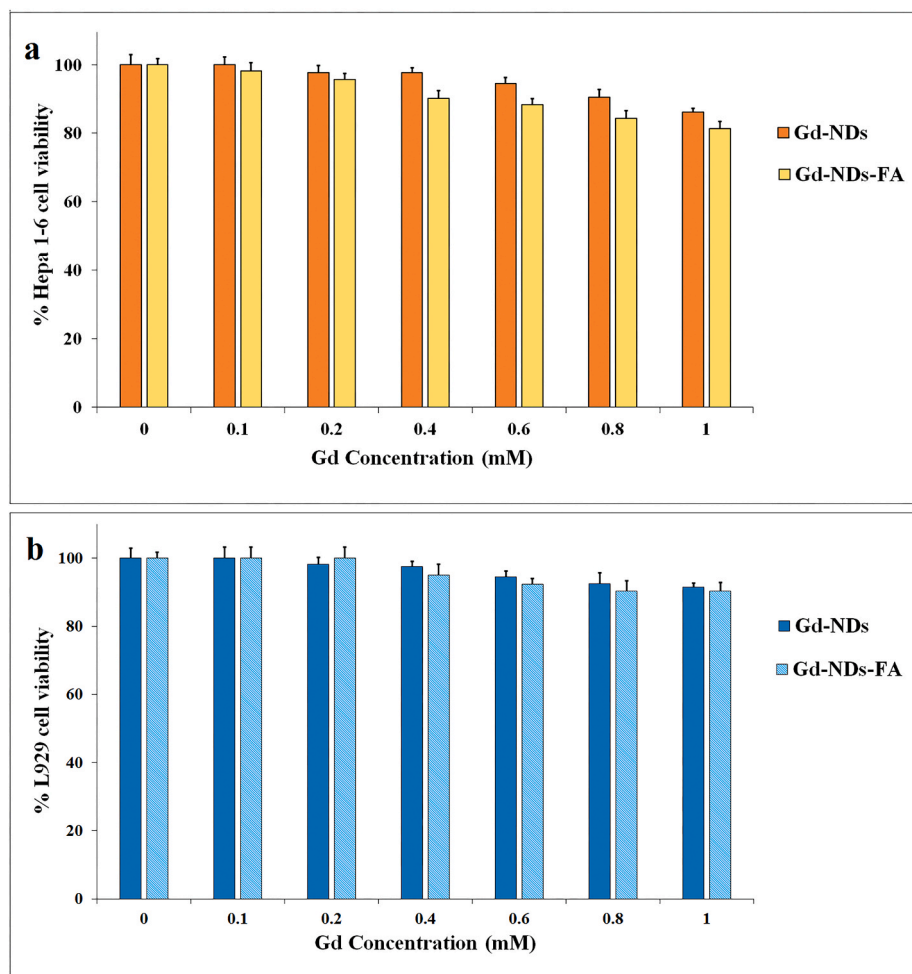


Fig. 4. Cell viabilities of Hepa1-6 cancer (a) and L929 normal (b) cells treated with Gd-NDs and Gd-NDs-FA at different Gd concentrations.

applications are bio-toxicity and side effects. Therefore, in this study, the *in vivo* potential toxicity of Gd/PFH@Alg-FA nanodroplets was evaluated by blood biochemistry (Fig. 5(a)) and histopathology (Fig. 5(b)) analysis on normal mice, 21 days after IV injection. As Fig. 5(a) shows, no significant differences were observed in the concentrations of AST, ALT, CREA, and BUN for the mice injected with Gd-loaded NDs compared to the control groups (p -value > 0.05). Furthermore, NDs did not exhibit any abnormality or organ damage in the histopathology examination of kidneys, liver, heart, lung, spleen, and brain compared to the control groups. These results support the high biocompatibility and biosafety of the fabricated nano-contrast agents for *in vivo* biomedical applications.

3.7. Hemolysis assay on human blood

The hemocompatibility of nanomaterials is a key factor for *in vivo* clinical applications since they can induce significant toxicity when interacting with human blood cells in the circulatory system [68]. When lanthanide elements (La, Gd) confront human RBCs, they induce hemolysis by creating pores in cell membranes [69]. In this study, hemolysis percentages caused by Gadovist, blank NDs, and Gd-NDs-FA with different Gd concentrations (0.07–10 mM) on fresh human blood for 3 h were evaluated (Fig. 6). Hemolytic activity of Gadovist and Gd-NDs-FA is increased with a Gd concentration-dependent trend so that the hemolysis caused by Gadovist and Gd-NDs-FA are 16.1% and 5.9%, respectively at 10 mM Gd. As Fig. 6 shows, the hemolytic activity of Gadovist in all concentrations is considerably higher than that of Gd-loaded NDs (p -value < 0.05). For concentrations up to 0.6 mM, the hemolytic activity of Gadovist is less than the acceptable standard limit of 6% [38], while by increasing its concentration to 10 mM, hemolysis reaches 16.1% probably due to the toxicity of gadolinium ions as a result of their interaction with RBCs, and damage to most of them [38]. Furthermore, blank NDs show no significant hemolytic activity at all concentrations. The hemolysis for Gd-loaded NDs at all used concentrations is less than 6%. Therefore, Gd-loaded NDs in the wide range of concentrations, including the range used in all experiments of this study

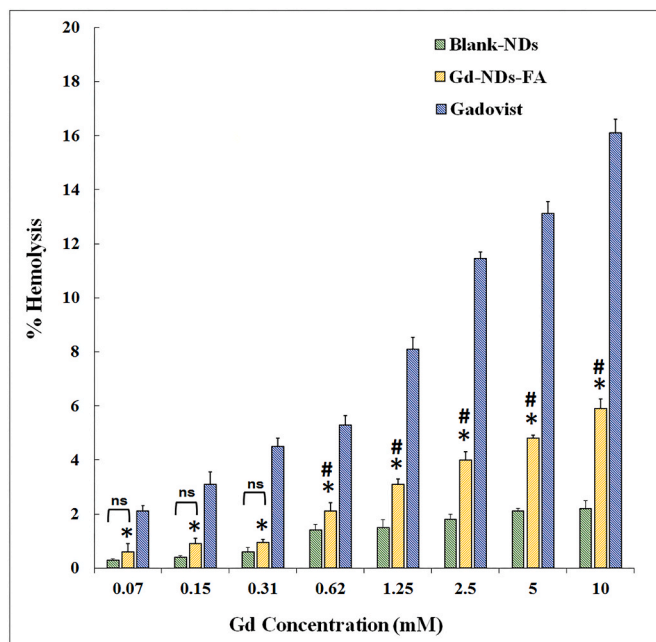


Fig. 6. Hemolytic activities of Gadovist, blank-NDs, and Gd-NDs-FA on human red blood cells (RBCs) at different Gd concentrations. Means $\pm$ SD ($n = 3$). Comparison with the Gadovist group, * p -value < 0.05 . Comparison with NDs-blank group, # p -value < 0.05 . ns, not significant. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(0.07–10 mM) are completely safe. The alginate shell as a biocompatible natural biopolymer decreases the hemolytic activity of Gd-NDs-FA. Therefore, loading of Gd^{3+} ions into alginate-stabilized NDs enhances their biosafety on human erythrocytes. The results suggest that synthesized Gd-loaded alginate stabilized NDs with low hemolytic activity,

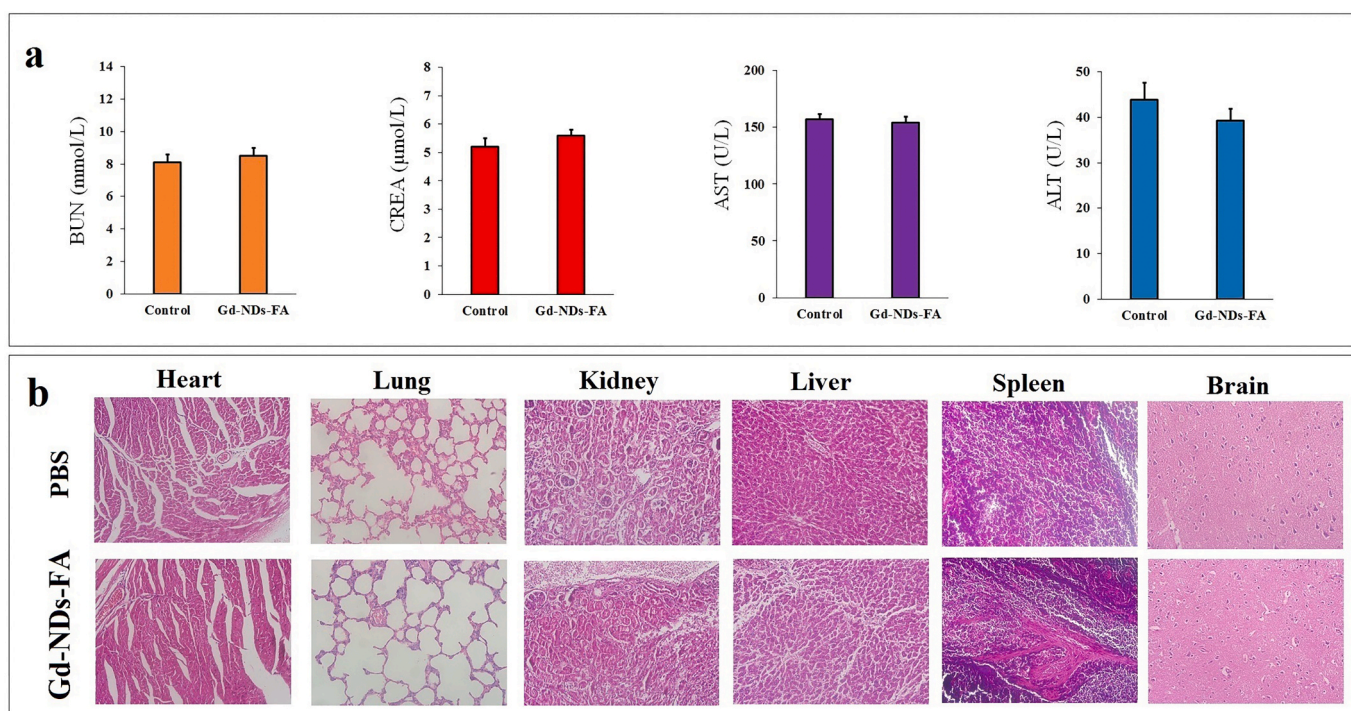


Fig. 5. (a) Blood biochemistry analysis and (b) Histological sections photographs of different organs (heart, kidney, spleen, liver, lung, and brain) of normal mice injected with PBS (as a control) and Gd-NDs-FA.

along with their good biocompatibility will be safe and non-toxic for *in vivo* biomedical applications.

3.8. Quantification of intracellular uptake

The intracellular uptakes of Gadovist and Gd-loaded NDs by Hepa1-6 cells were quantitatively assessed using ICP-OES after incubation for different times (2, 6, 12, 18, and 24 h). As Fig. 7 shows, intracellular uptakes of Gd^{3+} for all samples are increased with incubation time and reach the highest level after 24 h. The results demonstrate the significant role of surface modification of NDs with FA to enhance the cellular uptakes. For all incubation times, the internalization efficiencies of Gd-NDs-FA in Hepa1-6 cells are considerably higher than Gadovist and non-functionalized Gd-NDs (p-value < 0.01). This could be attributed to the overexpression of folate receptors on the Hepa1-6 cancer cell surface. Results also demonstrate that the difference between internalized concentrations of Gd^{3+} by targeted and non-targeted Gd-NDs reaches the maximum level after 12 h incubation, and Gd-NDs-FA exhibits approximately 2.5-fold higher intracellular uptake than Gd-NDs in this incubation time. There was no significant difference in the Gd-NDs-FA cellular uptake between 18, and 24 h incubation times (p-value > 0.05). The passive targeting effect becomes more important in longer incubation times. Therefore, the cellular uptake difference between targeted and non-targeted Gd-NDs at the final concentration of 0.8 mM Gd is decreased after 12 h.

3.9. In vitro MR imaging

To evaluate the potential application of synthesized Gd-loaded NDs as MRI contrast agents, L929 and Hepa1-6 cells were incubated with Gd-NDs and Gd-NDs-FA nanodroplets at different Gd^{3+} concentrations for 6 h, and they were scanned by the clinical MRI system. As Fig. 8 shows, enhancement of T_1 -weighted MRI signal intensities were observed for the Hepa1-6 cells incubated with Gd-loaded NDs compared to the control group so that enhancement of the signal intensities depends on the Gd^{3+} concentration, demonstrating the enhancement of the positive contrast. Furthermore, FA-functionalized Gd-NDs exhibited higher MRI signal enhancement than non-functionalized Gd-NDs for cancer cells (p-value < 0.05) so that the signal intensity in Hepa1-6 cells for Gd-NDs-FA

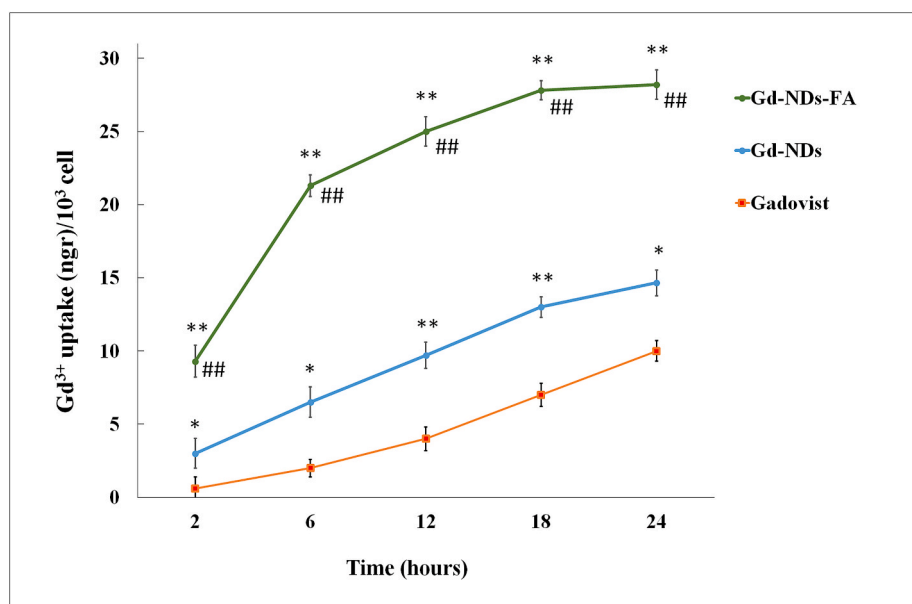


Fig. 7. Quantitative assessment of intracellular uptake of Gadovist, Gd-NDs, and Gd-NDs-FA by Hepa1-6 cancer cells after 2, 6, 12, 18, and 24 h incubation times by ICP-OES. Mean $\pm$ SD (n = 3). * p-value < 0.05 and ** p-value < 0.01, comparison with the Gadovist group. # p-value < 0.05 and ## p-value < 0.01, comparison with the Gd-NDs group.

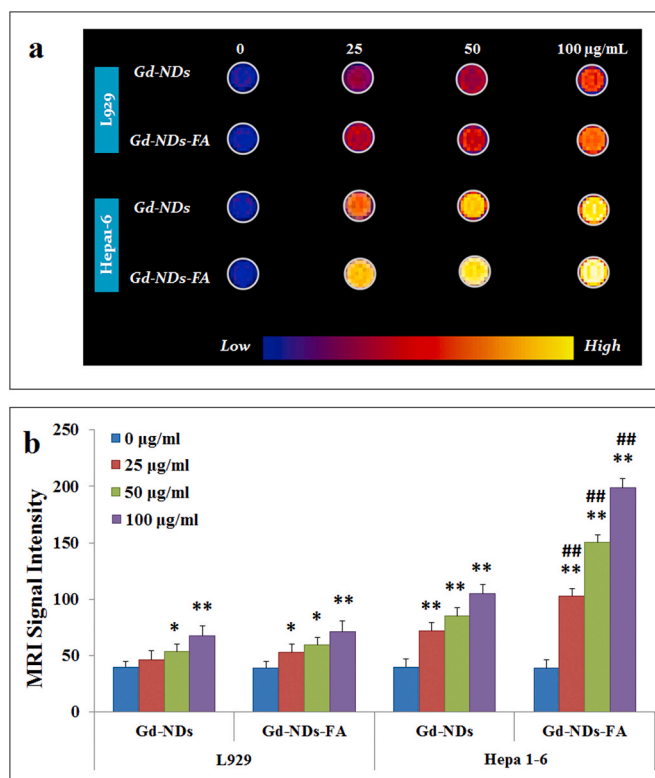


Fig. 8. (a) T_1 -weighted MRI images of Gd-NDs and Gd-NDs-FA in L929 and Hepa1-6 cells at different Gd^{3+} concentrations after 6 h incubation. (b) Signal intensity analysis for MRI images. Mean $\pm$ SD (n = 3). * p-value < 0.05 and ** p-value < 0.01, comparison with the control group. # p-value < 0.05 and ## p-value < 0.01, comparison between the Hepa1-6 and L929 cells treated with Gd-NDs-FA.

nanodroplets was approximately twice as much as that in Gd-NDs at 100 $\mu g\ mL^{-1}$ concentration. On the contrary, much less signal enhancement was observed for the L929 cells treated with both Gd-NDs formulations. The signal intensity of Hepa1-6 cells was approximately 3-fold higher

than the L929 group for Gd-NDs-FA at $100 \mu\text{g mL}^{-1}$ concentration. The MRI results demonstrated that Gd-NDs-FA nanodroplets could selectively enhance the T1-weighted MR image contrast in Hepa1-6 cells. These observations are further evidence for the targeting ability of NDs owing to overexpression of folate receptors on the cancer cells surface [54,70], which cause early diagnosis of cancers.

3.10. *In vivo* ultrasound imaging

To evaluate the *in vivo* capability of Gd-NDs-FA nanodroplets as an ultrasound contrast agent for tumor diagnosis, the Hepa1-6 tumor-bearing mice with intravenous injection of nanoprobes were used. Fig. 9 shows the ultrasound images of tumor before the nanodroplets injection and after 10, 20, and 60 min of the injection. Fig. 9(e) presents the signal intensity analysis in the tumor region. The hypoechoic area was found in the tumor before the Gd-NDs-FA injection (Fig. 9 (a)). The hyperechoic bright spots are visible in the tumor region 10 min after intravenous injection (Fig. 9 (b)), indicating phase transition of NDs and converting them to highly echogenic MBs in target tissue under ultrasound exposure. The brightness in the tumor is gradually enhanced with increasing time and reached the highest level after 60 min ($SI = 31.41 \pm 4.6$). This indicated that adequate NDs were accumulated into the tumor region and underwent the phase transition under ultrasound irradiation that significantly enhanced the ultrasound image contrast.

3.11. *In vivo* MRI

Although MR/US dual-modal imaging contrast agents have previously been reported, little attention has been paid to gadolinium-based

agents. The targeted MR imaging ability of synthesized Gd-NDs-FA nanodroplets in this study to diagnose cancer cells was further evaluated using the animal Hepa1-6 tumor model. Fig. 10 shows the coronal T₁-weighted MR images of the tumor-bearing mice before as well as 1, 6, and 24 h after intravenous injection of Gd-loaded NDs and signal intensity analyses of images in the tumor area. The results demonstrated that after injection of the Gd-NDs-FA emulsion, the signal intensities of the tumor were gradually increased. As Fig. 10 (a) shows, bright enhancement in the tumor can be clearly observed at 1 h after the injection. A significant signal enhancement ($SI = 148.87 \pm 12.64$) was observed in the tumor area 6 h after the injection. The signal intensity of the tumor at this time point was about 3-fold higher than that of pre-injection. The tumor signal intensity reduced to 86.44 ± 12.3 after 24 h of administration. Our results are in line with the study conducted by An et al. [71] that synthesized and evaluated hollow silica nanospheres (HSNSs) conjugated with Gd-DTPA and RGD ligand (HSNSs@ (DTPA-Gd)-RGD) as ultrasound and MRI dual-modal contrast agents. Also, MRI results are comparable with the study carried out by Morteza-zadeh et al. [37], who investigated gadolinium oxide nanoparticles and doxorubicin coated with β -cyclodextrin (CD)-based polyester and targeted by FA ($\text{Gd}_2\text{O}_3@ \text{PCD-FA-Dox}$) as T₁-weighted MRI contrast agent. In this study and our study, a significant amount of nanoprobes remains in the tumor region, even 6 h after intravenous injection, suggesting high *in vivo* stability of nanoprobes. The *in vivo* imaging results further confirmed that Gd-NDs-FA nanodroplets with appropriate relaxivity could be used to US/MR dual-modal imaging and they have specific targeting ability of cancer cells and selective accumulation in the tumor site, enabling early detection and accurate diagnosis of cancer.

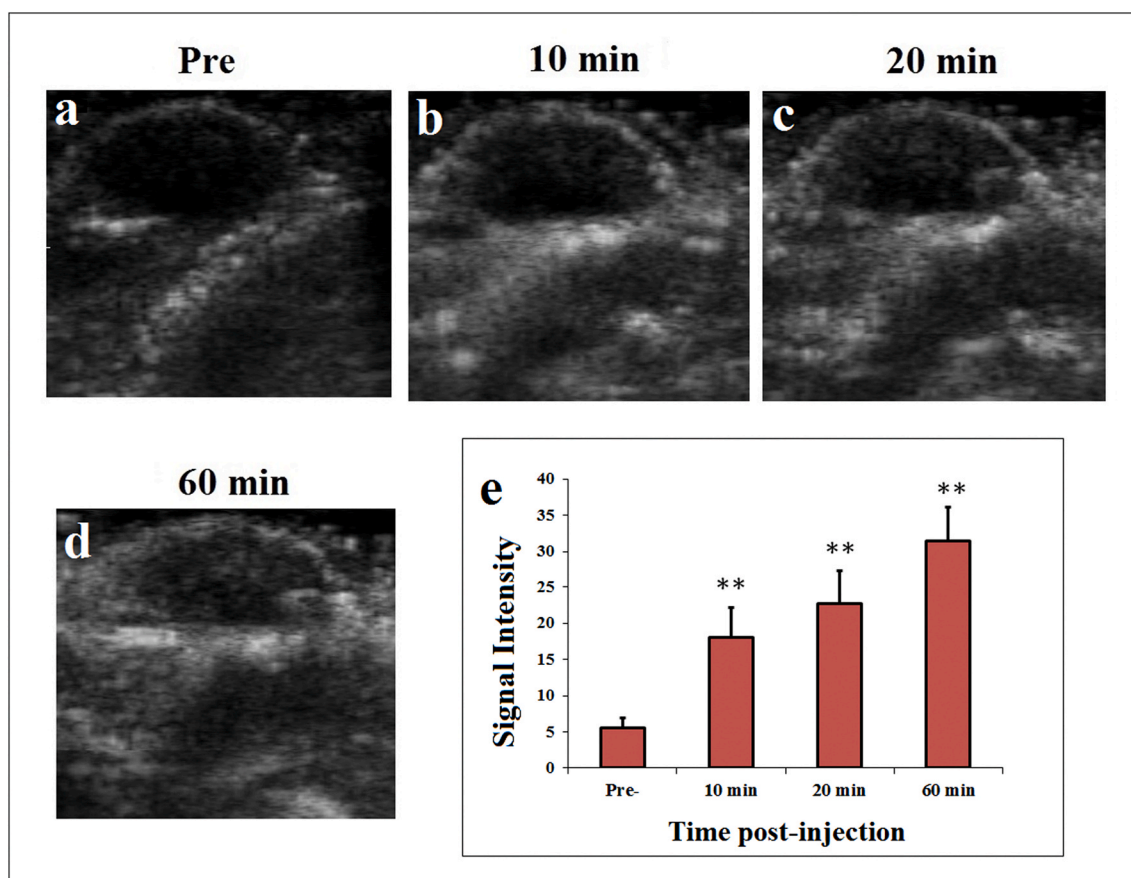


Fig. 9. *In vivo* ultrasound images of tumor of the mice bearing Hepa1-6 tumor (a) before and (b) 10 min, (c) 20 min, and (d) 60 min after intravenous injection of Gd-NDs-FA nanodroplets. (e) Signal intensities of ultrasound images in the tumor region. Mean $\pm$ SD ($n = 5$). * p -value < 0.05 and ** p -value < 0.01 , comparison with the pre-injection.

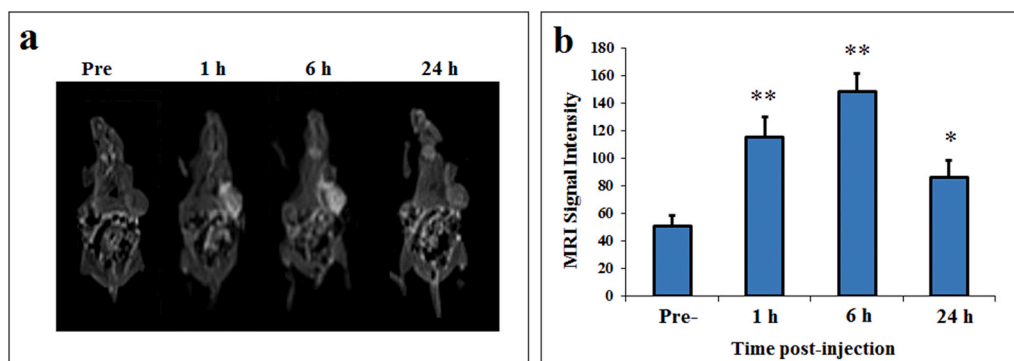


Fig. 10. (a) T₁-weighted MRI images before and 1, 6, and 24 h after injection of Gd-NDs-FA nanodroplets. (b) The signal intensity analysis for MR images in the tumor. Mean ± SD (n = 5). * p-value < 0.05 and ** p-value < 0.01, comparison with the pre-injection.

3.12. *In vivo* biodistribution of Gd-NDs-FA

The quantitative analysis of the Gd³⁺ content in the main organs can provide information about the biodistribution of contrast agents post-injection in the blood circulation which is important for *in vivo* biomedical applications [1]. The Gd³⁺ contents in the tumor and different organs of tumor-bearing mice after 24 h injection of Gd-NDs-FA are shown in Fig. 11. The highest Gd³⁺ concentration is found in the tumor, which indicates that Gd-NDs-FA nanodroplets are effectively accumulated in the tumor region and have a long circulation time. This can be attributed to the specific conditions of the tumor and the characteristics of the synthesized nanodroplets including their small size, alginate stabilizing shell, and targeting with folic acid. After the tumor, the highest Gd³⁺ concentrations were observed in the spleen and liver. It is predicted that the NDs were mainly accumulated in the reticuloendothelial system (RES) such as the spleen and liver and were metabolized by these organs. It is similar to the behavior of many nanomaterials *in vivo* in biomedicine applications [31,45,72]. The *in vivo* results demonstrated that the synthesized contrast agents in this study can effectively accumulate in the tumor region and provide high tumor image contrast and sufficient time for imaging that is the interest of recent research [1].

4. Conclusion

The development of multi-modal contrast agents combining diagnostic properties of two or more imaging techniques for the early detection of cancers has attracted considerable attention in recent years. In the present study, ultrasound-triggered phase transition Gd-loaded NDs were successfully prepared by the nano-emulsion process for dual-modal US/MR imaging. The synthesized NDs display favorable characteristics, such as small size, high stability, and controllable vaporization to echogenic MBs form. The small size and high stability of NDs cause them to be capable of enhancing circulation time and penetration into the tumor region, an important factor for molecular imaging. The results show that Gd-NDs significantly enhance ultrasound and MRI signal intensities with a high r_1 relaxivity of $6.40 \text{ mM}^{-1}\text{S}^{-1}$, which is even higher than that of the commercially clinical contrast agent (Gadovist). *In vivo* and *in vitro* toxicity assays reveal that developed Gd-loaded NDs possess excellent biocompatibilities. The functionalization of NDs with FA enhances their internalization into cancer cells via the active targeting strategy so that the cellular uptake of Gd-NDs-FA, after 12 h incubation, is approximately 2.5-fold higher than that of non-functionalized Gd-NDs. The *in vivo* experiments results further confirmed that Gd-NDs-FA nanodroplets have specific targeting ability of cancer cells and selective accumulation in the tumor region. In summary, Gd/PFH@Alg-FA nanodroplets are highly promising CAs for the US and MR molecular imaging of cancer, and they can provide complementary diagnostic information.

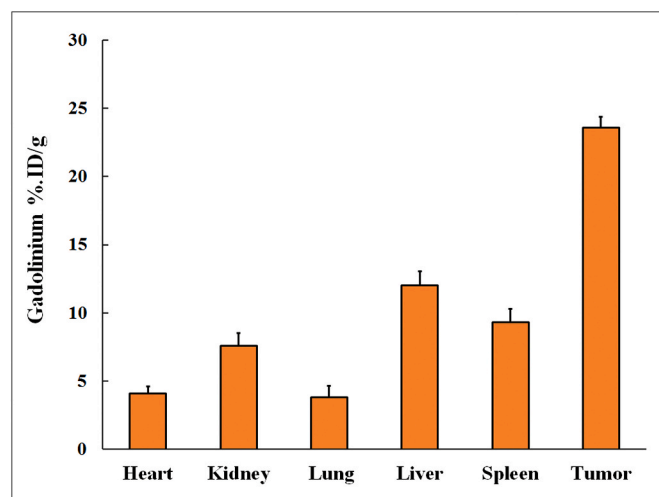


Fig. 11. Quantitative analysis of Gd³⁺ in tumor and major organs (liver, kidney, heart, spleen, and lung) of Hepa1-6 tumor-bearing mice, 24 h after intravenous injection of Gd-NDs-FA nanodroplets.

Author contributions

Fatemeh Maghsoudinia: Designed and performed experiments, Writing – original draft, Investigation. Mohamad Bagher Tavakoli: Supervision, Project administration. Roghaye Kamran Samani: Provided technical assistance, Developed the theory, Co-wrote the paper. Seyed Hossein Hejazi: Advisor, Reviewed and commented on the manuscript. Tayebe Sobhani: performing *in-vivo* experiments. Fatemeh Mehradnia: Development the methodology. Masoud A. Mehrgardi: Conceived of the presented idea, Supervision, Project administration, Reviewing, and Editing the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.talanta.2021.122245>.

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